Software Requirements Specification

Application for statistical analysis of exchange rates data

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1. Introduction

1.1. Purpose

The purpose of this document is to describe the features and requirements of a console application project. It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate and how the system will react to user activity. The application is intended to read data from the NBP API and conduct advanced statistical analysis based on the user's needs.

1.2. Product Scope

The system contains integration with the NBP API, statistical analysis algorithms used to process the data obtained from the API and a CLI for user interaction and display of the results.

2. Overall Description

2.1. User Needs

The user needs a system that can automatically process data describing exchange rates. It has to perform calculations that result in statistical measures at specific time intervals specified by the user. The statistics should be possible to display and export.

2.2. Assumptions and Dependencies

Dependencies:

A stable connection with the internet.
A stable connection to the NBP API service.
The data is obtained in the correct format.

Assumptions:

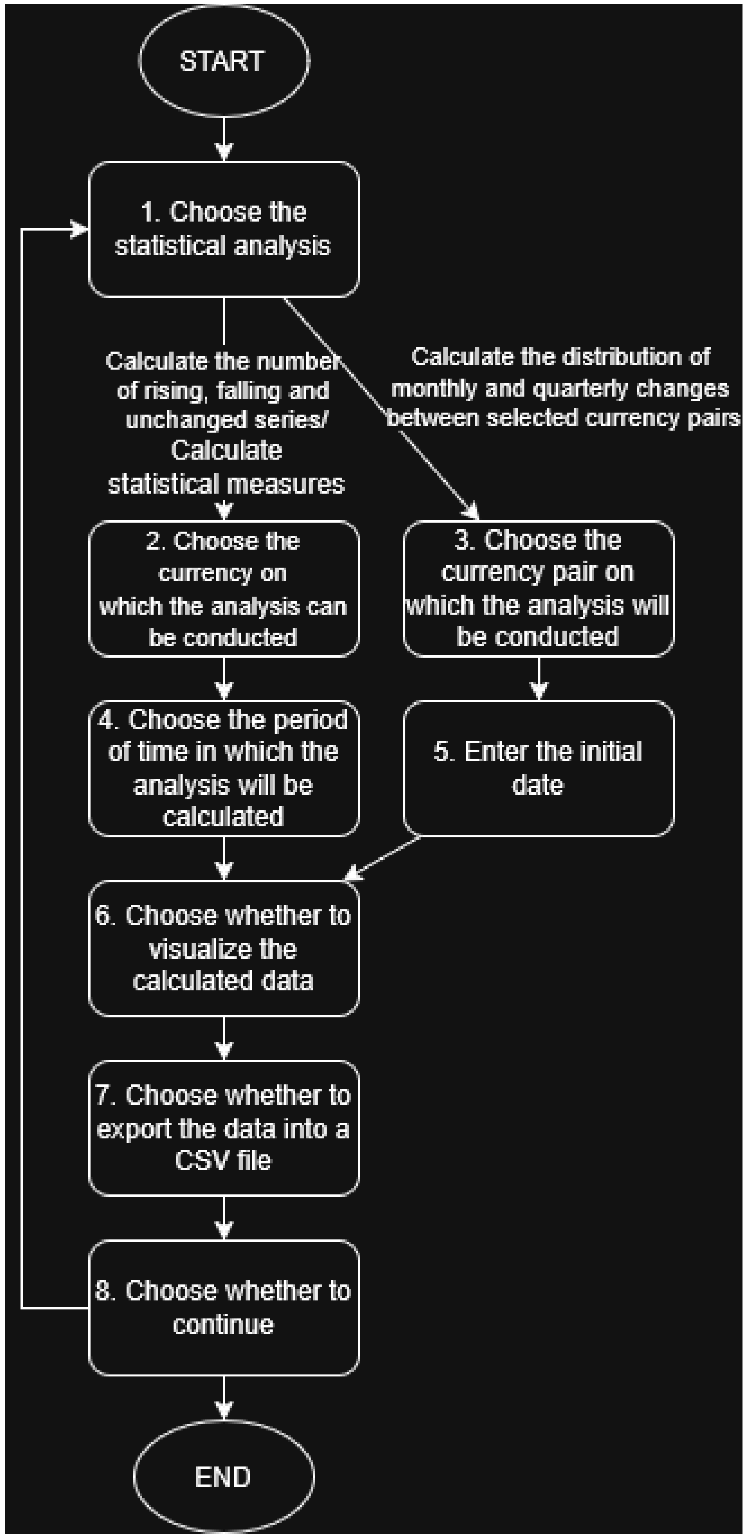
The NBP API is accessible and returns factual exchange rates data in the JSON/XML format

3. System Features and Requirements

3.1. Functional Requirements

1. The system describes its functionality to the user.
2. The user can choose between different statistical analyses.
3. The user should be able to visualize the calculated results in the form of a graph or data table (based on the kind of analysis).
4. The user should be able to export the results into a CSV file.
5. The user can define what currency should be used for the analyses.
6. The user (in analyses that use periods of time as a measurement) can define the period of time in which the analysis should be conducted. The user can choose between the periods of 1 week, 2 weeks, 1 month, 1 quarter, half a year and 1 year.
7. The user can choose to calculate the number of rising, falling and unchanged series in the selected currency and period.
8. The user can choose to calculate statistical measures such as median, mode, standard deviation and coefficient of variation in the selected currency and period.
9. The user can choose to calculate the distribution of monthly and quarterly changes between user selected currency pairs. The result is returned as a histogram of the frequency of value changes within the chosen period of time (displayed either as a graph or text).
10. The structure of the CLI is defined by the diagram shown below:

    The diagram shows every screen of the CLI as a block with a description. We also assume that in every screen there is an option to return to the previous screen and that in the event of an input error the CLI shows an error message before returning back to the last choice screen.



Explanation of blocks used in the diagram:

1. Choose the statistical analysis

   The system displays the main menu, allowing the user to select one of three available statistical analyses or exit the application. Each option is clearly described so the user understands what kind of calculation will be performed. The selection is entered by typing the chosen option in the console.

   For example:

   === NBP CURRENCY ANALYSIS SYSTEM ===

   Choose analysis type:
   1. Calculate the number of rising, falling, and unchanged sessions
   2. Calculate statistical measures (Median, Mode, Standard deviation, Coefficient of variation)
   3. Determine distribution of monthly and quarterly changes between currency pairs
   4. Exit application

   Type 1-4 and press Enter to choose the corresponding option:

2. Choose the currency on which the analysis can be conducted

   The system displays the available currencies and allows the user to choose the currency on which the previously chosen analysis will be conducted. The user can also return to the previous screen to change the analysis choice. The selection is entered by typing the chosen option in the console.

   For example:

=== SELECT CURRENCY ===

Here are the available currencies on which the analysis can be conducted:
1. USD
2. EUR
3. PLN
4. Return to previous selection

Type 1-4 and press Enter to choose the corresponding option:

3. Choose the currency pair on which the analysis will be conducted

    The system displays the available currencies and allows the user to choose the currency on which the analysis will be conducted similar to screen nr. 2. After the first choice the user is presented with another screen to choose the second currency. The screen is relatively unchanged compared to the previous one.

    For example:
    === SELECT CURRENCY ===

    Here are the available currencies on which the analysis can be conducted, choose the first/second (changes depending on the screen) currency:
    1. USD
    2. EUR
    3. PLN
    4. Return to previous selection

    Type 1-4 and press Enter to choose the corresponding option:

4. Choose the period of time in which the analysis will be calculated

    The system displays the lengths of time in which the analysis can be conducted and allows the user to choose one. The user can also return to the previous screen. The selection is entered by typing the chosen option in the console.

    For example:

    === SELECT TIME PERIOD FOR ANALYSIS ===

    Choose the length of the period for which the analysis will be conducted:

    1. 1 week
    2. 2 weeks
    3. 1 month
    4. 1 quarter (3 months)
    5. 1/2 year (6 months)
    6. 1 year
    7. Return to previous selection

    Type 1-7 and press Enter to choose the corresponding option:

5. Enter the initial date

    The system asks the user to enter the starting date for the analysis. It also describes the correct format that should be used to enter the date. The user can also return to the previous screen. The date is entered by typing it in the console using the correct format.

    For example:

    === DEFINE START DATE ===

Enter the start date of the analysis, use the format YYYY-MM-DD (e.g., 2026-01-15), if you wish to return to the previous choice enter R instead:

6. Choose whether to visualize the calculated data

   The system asks the user whether they want to display the data on the screen, either in the form of a graph or a table depending on the selected analysis type. The user can also return to the previous screen. The choice is entered by typing YES or NO into the console.

   For example:
   === SESSION ANALYSIS RESULT ===

   Do you want to display the calculated data? Type YES or NO to confirm. If you wish to return to the previous choice enter R instead:

7. Choose whether to export the data into a CSV file

   The system asks the user whether they want to export the data into a CSV file. The user can also return to the previous screen. The choice is entered by typing YES or NO into the console.

   For example:
   === EXPORT ANALYSIS RESULT ===

   Do you want to export the calculated data into a CSV file? Type YES or NO to confirm. If you wish to return to the previous choice enter R instead:

8. Choose whether to continue

   The system asks the user whether they wish to continue using the application. The choice is entered by typing YES or NO into the console. If the user chooses to continue they are shown the main screen, if they choose to finish using the application it shuts down.

For example:
=== CONTINUE===

Do you want to continue using the application? Type YES to return to the main screen or NO to shut down the application:

3.2. Non-Functional Requirements

3.2.1. Performance

The system should process the users request in under 5 seconds (assuming the API response is under 2 seconds).

3.2.2. Security

The application does not store any vulnerable user data. The connection with the NBP API is carried out under the HTTPS protocol.

3.2.3. Usability

The CLI should describe its functionality to the user.
The CLI should give feedback depending on the users actions.
The system should notify the user of downtime by displaying an appropriate error message.

3.2.4. Reliability

System downtime can only be caused by the downtime of the external NBP API.
The system should handle API connection failure (either due to API downtime or loss of internet connection).
The system should recover from failure within 1 minute of reestablishing connection with the API.

External Interface Requirements

Software Interfaces: Integration with the NBP API

User Interface: Communication with the system using a CLI

4. Other Requirements

4.1. Legal and Regulatory Requirements

The system shall indicate that NBP is the source of the data used for statistical analysis.
The data is for informational use only and is not legally binding.

4.2. Risk Management (FMEA Matrix)

API failure	Cannot fetch exchange rate data	8	NBP API downtime / server issues	5	3	120	Retry logic, fallback message, notify user
Internet connection failure	Application cannot communicate with API	7	Local network issues	6	4	168	Detect offline, retry automatically, alert user
Application crash	Loss of current session / user data	9	Unhandled exceptions, bugs	4	5	180	Error handling, logging, input validation, auto-restart
Invalid user input	Incorrect or failed analysis	6	Typo, unsupported currency code, invalid date range	7	4	168	Input validation, prompt user with correct format
Statistical analysis algorithm error	Wrong results calculated	8	Bug in algorithm, edge case not handled	3	6	144	Unit tests, code review, validation with sample data
CSV export fails	Results not saved or lost	7	File permission issues, disk full, incorrect path	5	5	175	Check file paths, handle exceptions, confirm write success
Graph visualization fails	User cannot view visual output	5	Missing library,	4	6	120	Validate data, check dependencies,

			incorrect data format				fallback to text table

5. Appendices

5.1. Glossary

NBP – Narodowy Bank Polski

API – Application Programming Interface

CLI – Command-Line Interface

JSON – JavaScript Object Notation

XML – Extensible Markup Language

CSV – Comma-Separated Values

5.2. Use Cases and Diagrams

5.2.1. Select Analysis Parameters

Brief Description

This use case allows the user to define the parameters required to perform a statistical analysis of currency exchange rates.

Initial Step-by-Step Description

5.2.1.1. The user starts the system.

5.2.1.2. The system prompts the user to choose the type of analysis.

5.2.1.3. The user selects the analysis type.

5.2.1.4. The system asks the user to select a currency or currency pair.

5.2.1.5. The user selects the desired currency or currency pair.

5.2.1.6. The system prompts the user to select a time period. The user can choose one of the available options: 1 week, 2 weeks, 1 month, 1 quarter, half a year, or 1 year.

5.2.1.7. The user selects the time period.

5.2.1.8. The system validates the provided parameters.

5.2.1.9. The system confirms the selected parameters.

5.2.2. Analyze Currency Session Trends

Brief Description

This use case calculates the number of upward, downward, and unchanged sessions for a selected currency over a specified period.

Initial Step-by-Step Description

5.2.2.1. The user initiates the session trend analysis.
5.2.2.2. The system invokes the parameter selection process.
5.2.2.3. The system retrieves exchange rate data.
5.2.2.4. The system compares each session with the previous one.
5.2.2.5. The system classifies sessions as upward, downward, or unchanged.
5.2.2.6. The system counts the number of each type of session.
5.2.2.7. The system prepares the results.
5.2.2.8. The system displays the results to the user.

5.2.3. Calculate Statistical Measures

Brief Description

This use case calculates statistical measures such as median, mode, standard deviation, and coefficient of variation for a selected currency.

Initial Step-by-Step Description

5.2.3.1. The user selects the statistical analysis option.
5.2.3.2. The user selects the statistical analysis option.
5.2.3.3. The system invokes the parameter selection process.
5.2.3.4. The system retrieves exchange rate data.
5.2.3.5. The system extracts the relevant values for the selected period.
5.2.3.6. The system calculates: median, mode, standard deviation, coefficient of variation.

5.2.3.7. The system prepares the computed statistics.
5.2.3.8. The system displays the results to the user.

5.2.4. Analyze Currency Change Distribution

Brief Description

This use case analyzes the distribution of currency changes over time and presents the results.

Initial Step-by-Step Description

5.2.4.1. The user selects the change distribution analysis.
5.2.4.2. The user selects the change distribution analysis.
5.2.4.3. The system invokes the parameter selection process.
5.2.4.4. The user selects a currency pair.
5.2.4.5. The system retrieves exchange rate data.
5.2.4.6. The system calculates daily changes between consecutive sessions.
5.2.4.7. The system aggregates changes into monthly or quarterly periods.
5.2.4.8. The system groups the results into value ranges (bins).
5.2.4.9. The system generates the histogram.
5.2.4.10. The system displays the distribution to the user.

5.2.5. Retrieve Exchange Rate Data

Brief Description

This use case retrieves currency exchange rate data from the external NBP API.

Initial Step-by-Step Description

5.2.5.1. The system prepares a request to the NBP API.
5.2.5.2. The system sends the request.
5.2.5.3. The NBP API processes the request.
5.2.5.4. The NBP API returns exchange rate data.
5.2.5.5. The system receives the response.
5.2.5.6. The system validates the received data.
5.2.5.7. The system stores the data temporarily for further processing.

5.2.6. View Analysis Results

Brief Description

This use case presents the results of the performed analysis to the user in a readable format.

Initial Step-by-Step Description

5.2.6.1. The system receives processed analysis results.
5.2.6.2. The system formats the results.
5.2.6.3. The system selects an appropriate presentation format (e.g., table, text, histogram).
5.2.6.4. The system displays the results to the user.

5.2.7. Export Results

Brief Description

This use case allows the user to export analysis results to an external file.

Initial Step-by-Step Description

5.2.7.1. The user selects the export option.
5.2.7.2. The system prepares the data for export.
5.2.7.3. The system generates the file.
5.2.7.4. The system saves the file to the specified location.
5.2.7.5. The system confirms successful export.